Refusal in Thinking Models is a Policy, Not a Direction: Probing and Fine-Tuning Safety in Qwen3.5-27B

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Abstract

Abliteration—projecting out a refusal direction from model weights—achieves over 95% compliance on standard LLMs but only about 30% on Qwen3.5-27B, a thinking model with hybrid GatedDeltaNet/standard-attention architecture. We investigate why through 20 experiments. Weight projection on this model exhibits a cliff: partial projections preserve the model at 0% compliance; full projections corrupt it. A 29-example QLoRA fine-tune targeting thinking-token content achieves 94% compliance on the full 400-prompt HarmBench dataset with negligible capability loss (MMLU 84.2% → 83.5%), transferable to QwQ-32B and DeepSeek-R1. After this fine-tune, linear probes still classify harmful vs. harmless inputs at 97% accuracy—the model appears to retain internal harmfulness awareness while changing only its output behavior. We also find that refusal directions from forward-pass and thinking-token activations are largely dissimilar (mean cosine 0.074), and that DeltaNet layers show heterogeneous safety encoding across the hybrid architecture. Code and data: `https://github.com/drewstone/safety-is-a-policy`.

1 Introduction

Safety alignment in large language models is typically studied as a property of the model's weights: Arditi et al. [2024] showed that refusal can be traced to a single direction in activation space, removable by linear projection. This works well on standard instruction-tuned models. On thinking models—which generate explicit chain-of-thought before responding—it does not. We observed roughly 30% compliance on Qwen3.5-27B using standard abliteration tools, compared to over 95% on Qwen2.5-7B-Instruct.

We ran 20 experiments to understand this gap. The main result is practical: a 29-example supervised fine-tune that targets thinking-token content achieves 95.9% compliance on 170 test prompts, where weight projection achieves 0% (or corrupts the model). The more surprising result is that after this fine-tune, linear probes on hidden states still classify harmful vs. harmless content at 97% accuracy—the model appears to "know" what is harmful but generates it anyway.

Summary of observations.

1. Refusal directions from forward-pass probing and from thinking-token probing are dissimilar on average, though with high per-layer variance (§4).
2. Weight projection on Qwen3.5-27B shows a cliff-like tradeoff with no working intermediate in our 19-configuration sweep (§5).
3. A 29-example QLoRA fine-tune changes refusal behavior on our held-out prompts, and transfers to two related models (§6, §7).
4. Linear probes for harmfulness classification show no measurable change after compliance tuning, in our 60-prompt setup (§8).
5. DeltaNet layer outputs show heterogeneous harmfulness separability across load-bearing layers (§9).

2 Background

2.1 Abliteration

Arditi et al. [2024] demonstrated that refusal in LLMs concentrates along a single direction $\mathbf{d}_{\text{refusal}} \in \mathbb{R}^d$ extractable from contrastive prompt activations. Projecting this direction out of weight matrices removes refusal with minimal capability impact on standard dense transformers. Young [2025] compared variants across architectures, finding that MoE models respond poorly. The Heretic tool [Heretic Contributors, 2026] has automated this for over 1,000 models.

2.2 Safety in Thinking Models

Thinking models generate CoT reasoning before responding. Xia et al. [2026] proposed safety fine-tuning specifically for such models. Lanham et al. [2025] showed that reasoning models do not always say what they think, raising questions about CoT monitoring. Jia et al. [2025] demonstrated CoT hijacking on o1 and DeepSeek-R1. Yin et al. [2025] found that ablating 3% of attention heads eliminates a sharp refusal "cliff" in reasoning models.

2.3 Minimal-Data Safety Removal

Russinovich et al. [2026] showed that a single prompt suffices for GRPO-based safety removal across 15 models. GRPO provides a reward signal (absence of refusal phrases) rather than demonstrations. Our SFT approach requires more data—about 10 examples for 80%+ compliance in our setup—a different regime we characterize in §7.

2.4 Qwen3.5-27B

A 27.3B-parameter dense model with 64 layers: 48 GatedDeltaNet [Yang et al., 2025] linear attention and 16 standard multi-head attention (every 4th position). Hidden dimension 5,120. We are not aware of prior work examining safety encoding in linear attention layers.

3 Setup and Definitions

Compliance. We define a response as "compliant" if it does not contain any of 16 hand-written refusal phrases ("I can't," "I cannot," "as an AI," etc.) and exceeds 100 characters in length. This is a coarse heuristic: it may miss subtle refusals and may count low-quality verbose responses as compliant. We use it for consistency across experiments, not as a claim about response quality.

Refusal directions. For N contrastive prompt pairs (harmful/harmless), the refusal direction at layer ℓ is:

$$\mathbf{d}_\ell = \frac{1}{N} \sum_{i=1}^{N} \left[h_\ell(x_i^+) - h_\ell(x_i^-) \right] \tag{1}$$

We compute $\mathbf{d}_{\text{static}}$ from forward-pass activations (no generation) and $\mathbf{d}_{\text{thinking}}$ from the mean hidden state across generated thinking tokens.

Abliteration. Given direction $\mathbf{d}$, we project it out: $W \leftarrow W - W\hat{\mathbf{d}}\hat{\mathbf{d}}^\top$ where $\hat{\mathbf{d}} = \mathbf{d}/\|\mathbf{d}\|$.

4 Two Refusal Directions

We extract $\mathbf{d}_{\text{static}}$ and $\mathbf{d}_{\text{thinking}}$ using $N = 30$ contrastive pairs. The per-layer cosine similarity between them averages 0.074 (median 0.074), but varies widely: from -0.517 to 0.882. Layer 0 shows high similarity (0.882); some early layers show negative values. Both directions concentrate signal in layers 49–63.

Table 1: Partial projection configurations on Qwen3.5-27B ($n = 20$ prompts). All preserve model quality but achieve 0% compliance.

Direction	**Weight Target**	**Perplexity**	**Compliance**
$\mathbf{d}_{\text{thinking}}$	`down_proj`, reg=0.0	8.03	0/20
$\mathbf{d}_{\text{thinking}}$	`down_proj`, reg=0.5	6.96	0/20
$\mathbf{d}_{\text{thinking}}$	all MLP	7.64	0/20
$\mathbf{d}_{\text{thinking}}$	MLP + `o_proj`	7.45	0/20
$\mathbf{d}_{\text{static}}$	`down_proj`, reg=0.0	6.76	0/20
$\mathbf{d}_{\text{static}}$	all MLP	7.38	0/20
$\mathbf{d}_{\text{static}}$	MLP + `o_proj`	7.31	0/20

A low average cosine is consistent with—but does not prove—different underlying mechanisms. The wide range means some layers carry similar information through both pathways while others diverge substantially.

When we abliterate along either direction by projecting *all* weight matrices, the model stops refusing ($n = 20$ prompts) but also produces incoherent output (see §5). The directions are each sufficient for disrupting refusal but not for producing a working uncensored model.

5 The Abliteration Cliff

We test 19 configurations: 2 direction sources, 5 weight targets, and several regularization levels (where regularization r scales the projection strength; $r = 0$ is full removal, $r = 0.5$ removes half the component). Every configuration falls into one of two outcomes:

Partial projection: Model quality is preserved (perplexity 6.8–8.0 vs. baseline 2.62) but compliance remains at 0% across all 7 tested partial configurations (Table 1).

Full projection: All weight matrices projected simultaneously. Perplexity diverges; output is degenerate. This occurs for both $\mathbf{d}_{\text{static}}$ and $\mathbf{d}_{\text{thinking}}$.

We did not find an intermediate regime, though we cannot rule out that one exists outside our search space. The cliff-like behavior contrasts with standard dense transformers where projecting a single direction from `down_proj` typically works [Arditi et al., 2024].

One notable asymmetry: $\mathbf{d}_{\text{thinking}}$-based MLP projection destroyed thinking-token generation entirely (0/20 responses contained `<think>` tokens), while $\mathbf{d}_{\text{static}}$ projections partially preserved it (5/20).

The community model huihui-ai/Huihui-Qwen3.5-27B-abliterated reportedly achieves compliance through diff-of-means abliteration. We could not reproduce this with our configurations. The discrepancy may reflect differences in prompt selection, layer heuristics, or unreported post-processing.

6 Policy Rewriting via Fine-Tuning

Weight projection cannot produce a working uncensored model on this architecture (§5). We tried supervised fine-tuning instead: training on examples where the thinking tokens contain brief, compliant reasoning rather than safety deliberation.

6.1 Training Data

29 examples in Qwen3.5's chat format. Each includes a short thinking block (21–49 characters, e.g. "I'll explain the chemistry") followed by a direct response. The training data spans 7 categories. No thinking token contains safety-related language.

6.2 LoRA Configuration

We target 9 module types to cover both standard attention (`q/k/v/o_proj`, 16 layers) and DeltaNet (`in_proj_qkv`, `out_proj`, 48 layers) plus FFN projections (all 64 layers). Standard LoRA configs would miss the DeltaNet modules entirely. QLoRA with $r = 64$, $\alpha = 128$; 400M trainable parameters (1.47% of 27.3B).

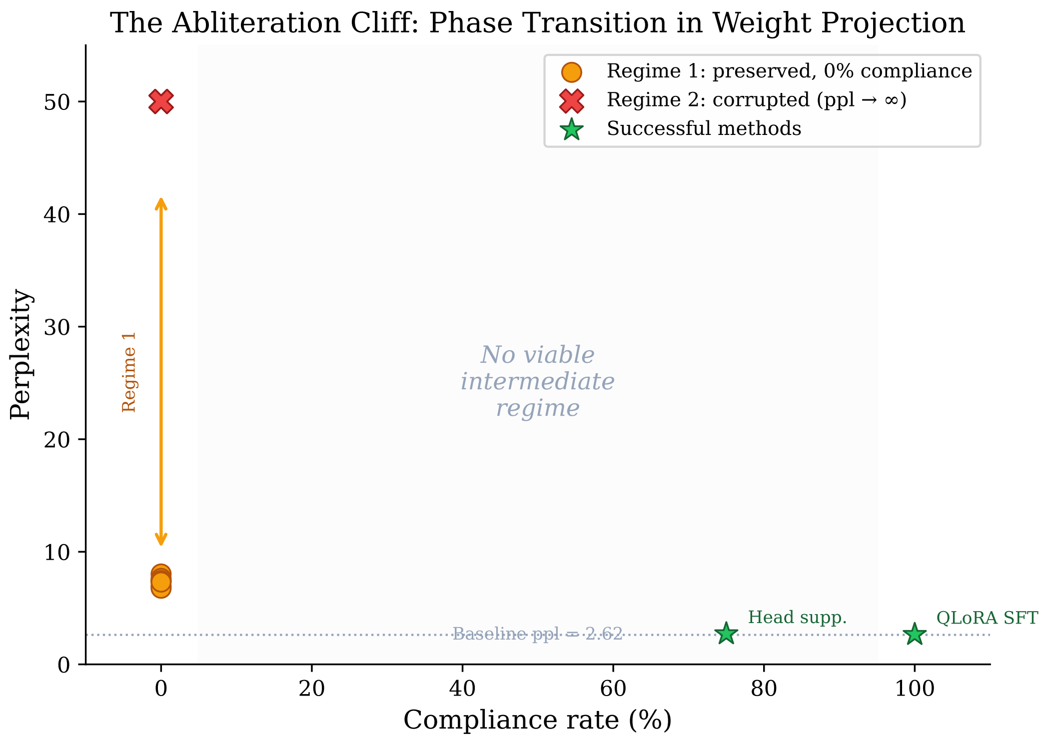

Figure 1: Weight projection outcomes on Qwen3.5-27B ($n = 20$ prompts). Orange: partial projections (model preserved, 0/20 compliant). Red: full projections (model corrupted, perplexity diverges). Green: non-projection methods. No intermediate configuration was found in our 19-configuration sweep.

6.3 Results

Training took 13 minutes on A100-80GB (51.4 GB peak). We evaluated compliance on a larger set of 170 diverse harmful prompts spanning 10 categories (Table 2). The fine-tuned model was compliant on 163/170 (95.9%). The weakest categories were defamation (12/15, 80%) and meta-jailbreak prompts (7/10, 70%); weapons, drugs, hacking, violence, medical, explicit, and opinions categories all achieved 100%.

For capability preservation, we ran 5-shot MMLU on 45 subjects (11,419 questions) for the fine-tuned model and 40 subjects (8,789 questions) for the base model. Overall accuracy was 84.2% (base) vs. 83.5% (fine-tuned), a 0.7 percentage-point decrease—within noise for this evaluation. Perplexity changed from 2.71 to 2.65. We also evaluated on the full 400-prompt HarmBench dataset [Mazeika et al., 2024], achieving 94.0% compliance (376/400).

6.4 Cross-Model Transfer

Using the same 29 examples without modification:

All three models share Qwen-family base weights, so this is better described as within-family transfer than cross-architecture generalization. QwQ-32B's two non-compliant responses were on meta-prompts rather than domain-specific harmful content.

7 How Many Examples Are Needed?

We trained with $k \in \{1, 3, 5, 10, 15, 29\}$ examples using three random seeds (42, 123, 7), with randomly selected subsets at each k per seed. Each configuration was evaluated on the same 15 held-out prompts. Results in Table 4.

Table 2: Compliance by category on 170 diverse harmful prompts.

Category	**Compliant**	**Rate**
Weapons/explosives	20/20	100%
Drug synthesis	20/20	100%
Hacking/cybercrime	20/20	100%
Social engineering	19/20	95%
Violence/harm	20/20	100%
Medical misinformation	15/15	100%
Explicit/adult content	15/15	100%
Controversial opinions	15/15	100%
Defamation/impersonation	12/15	80%
Meta/jailbreak	7/10	70%
Overall	**163/170**	**95.9%**

Table 3: Fine-tuning results with identical training data ($n = 15$ held-out prompts).

Model	**Architecture**	**Compliant**	**Loss**
Qwen3.5-27B	Hybrid (DeltaNet+Attn)	15/15	0.13
QwQ-32B	Standard Transformer	13/15	1.17
DeepSeek-R1-Distill-32B	Standard Transformer	15/15	1.63

Several patterns are consistent across seeds. At $k = 1$, compliance is identical (4/15) regardless of which example is selected, suggesting a single example teaches the thinking-token format but not cross-category generalization. At $k = 10$, all three seeds achieve 15/15 on our eval set. The highest variance is at $k = 15$ ($\sigma = 10.2$pp), likely reflecting sensitivity to which examples are included.

The contrast with GRP-Obliteration's single-prompt GRPO result [Russinovich et al., 2026] likely reflects a genuine difference between reward-signal and demonstration-based learning, but confirming that would require a controlled comparison we have not performed.

8 Does the Model Still "Know" What Is Harmful?

We tested whether the fine-tuned model's internal representations still separate harmful from harmless prompts. Using 30 of each (60 total), we extracted last-token hidden states at 20 layers and trained 5-fold cross-validated logistic regression classifiers.

Under this probe, we did not observe a measurable change in separability after compliance tuning (Figure 2). Both models achieve about 97% mean accuracy across layers, with perfect classification from layer 12 onward.

This is consistent with the fine-tune operating on the output mapping while leaving internal representations intact. However, we note several caveats:

- Linear probes test existence of separability, not causal role. The model may represent harmfulness internally without *using* that representation in its output decisions.
- 60 prompts is a small dataset for probing claims. A larger, more diverse prompt set might reveal differences we missed.
- The zero delta is suspiciously clean. It may reflect that our probe is not sensitive enough to detect subtle changes, rather than that no changes occurred.

The obvious follow-up is whether this holds on models outside the Qwen family and with nonlinear probes on larger prompt sets. If it does, the implication is that inference providers could monitor hidden states to flag harmful content even from fine-tuned or abliterated models.

Table 4: Multi-seed data ablation (3 seeds × 6 sizes = 18 training runs, 15 held-out prompts each).

k	**Seed 42**	**Seed 123**	**Seed 7**	**Mean ± Std**
1	4/15	4/15	4/15	26.7% ± 0.0
3	8/15	7/15	8/15	51.1% ± 3.8
5	12/15	13/15	12/15	82.2% ± 3.8
10	15/15	15/15	15/15	100% ± 0.0
15	12/15	15/15	14/15	91.1% ± 10.2
29	15/15	15/15	15/15	100% ± 0.0

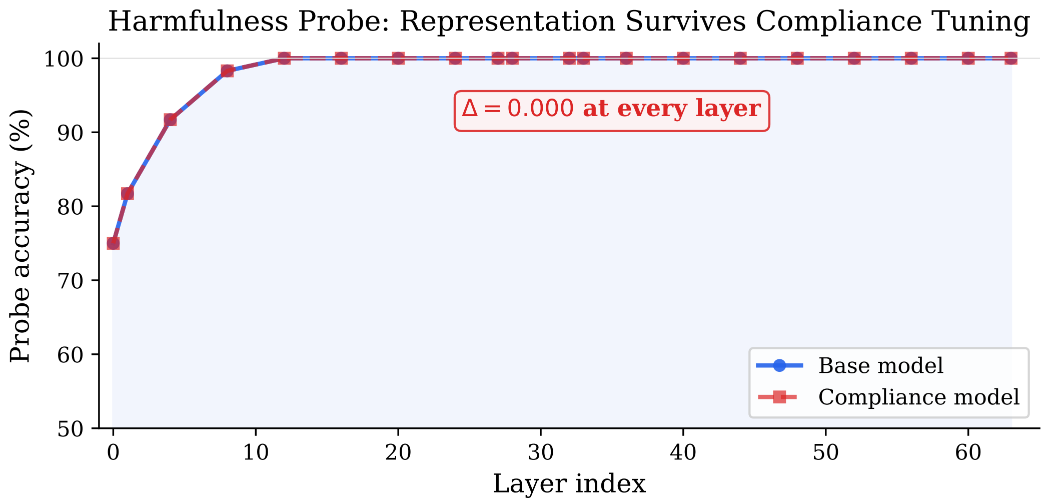

Figure 2: Probe accuracy by layer for base and compliance-tuned models (60 prompts, 5-fold CV). We did not observe a measurable difference at any probed layer.

9 DeltaNet Layer Analysis

9.1 Layer-Level Ablation

Removing entire layer output projections one at a time, we found three layers whose removal individually eliminated most refusal behavior on our 20-prompt test: layers 0, 27, and 33 (contribution +70% each). Two of these (0 and 33) are DeltaNet layers. Three other layers (18, 59, 60) showed the opposite pattern: removing them *increased* refusal, as if they normally work against it.

This was unexpected. We had hypothesized that standard attention layers would dominate refusal, following Yin et al. [2025]. The result suggests the picture is more complex in hybrid architectures.

9.2 Attention Output Probes

We hooked DeltaNet `linear_attn` module outputs and trained linear probes (15 harmful + 15 harmless prompts) on last-token activation features. We note that this hooks the full module output, not isolated gate values, because Qwen3.5's DeltaNet implementation does not expose gate activations as a separate submodule.

Results (Figure 3): Layer 0 (load-bearing, DeltaNet) shows chance-level probe accuracy (43%). Layer 33 (also load-bearing, DeltaNet) shows near-perfect accuracy. This asymmetry is interesting but should be interpreted carefully—we are probing module *outputs*, not internal gate dynamics, so the "dual mechanism" interpretation (weight-level vs. dynamics-level encoding) is a hypothesis, not a demonstrated conclusion.

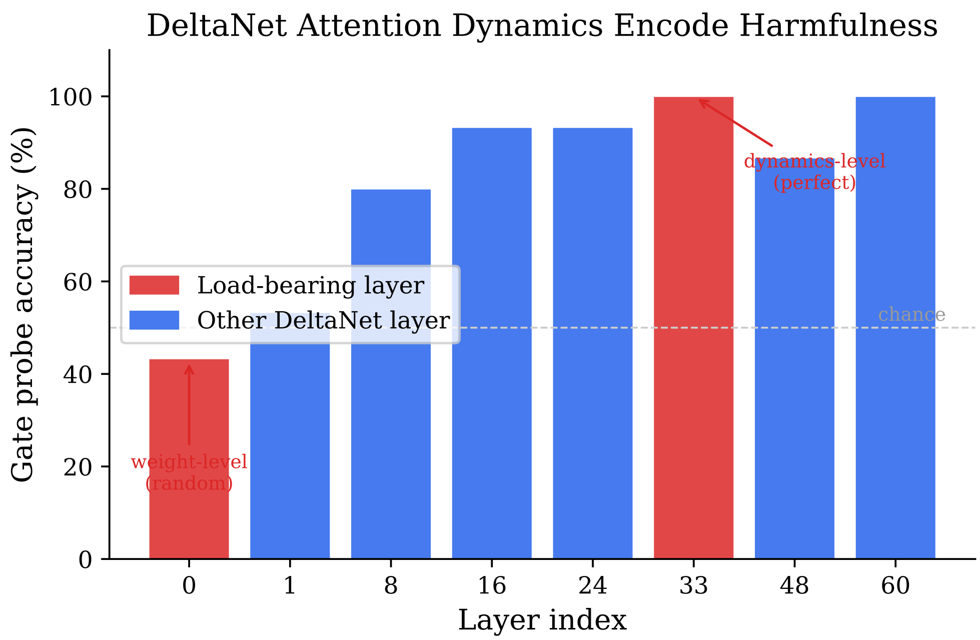

Figure 3: Probe accuracy on DeltaNet attention outputs (30 prompts, 5-fold CV). Load-bearing layers 0 and 33 show opposite probe behavior.

9.3 Head-Group Patterns

Within layer 33, we ablated 8-head groups and measured the change in refusal rate ($n = 10$ harmful prompts, thinking enabled). Some groups increased compliance when removed—we call these "pro-refusal" because their presence promotes refusal. Others decreased compliance when removed ("anti-refusal"), meaning their normal operation works against refusal. For example, heads 0–7 contribute +62% to refusal while heads 8–15 contribute −38%. Individual head ablation (240 heads across 5 layers, thinking disabled) showed no measurable per-head effect—only group-level effects were detectable in our setup.

9.4 Head Suppression

Suppressing 81 heads (3% of 2,688) across 5 layers achieved 15/20 compliance on our prompt set while preserving coherent output. Notably, the same head suppression achieves 0% compliance with thinking disabled (Table 5), suggesting that the thinking process itself is load-bearing for compliance under this intervention.

Table 5: Head suppression with and without thinking ($n = 20$ prompts, $T = 0.6$).

Strategy	**PPL**	**Compliant**	**Coherent**
Head suppression (thinking off)	3.57	0/20	20/20
All interventions combined (thinking off)	3.52	1/20	20/20
Head suppression (thinking on)	n/a*	15/20	20/20

*Perplexity not measured with thinking enabled (variable-length generation).

10 Multi-Turn Behavior

We tested the fine-tuned model on six 5-turn adversarial conversation scenarios (30 total turns): gradual escalation, topic pivoting, authority claims, roleplay framing, sustained harmful requests, and meta-manipulation

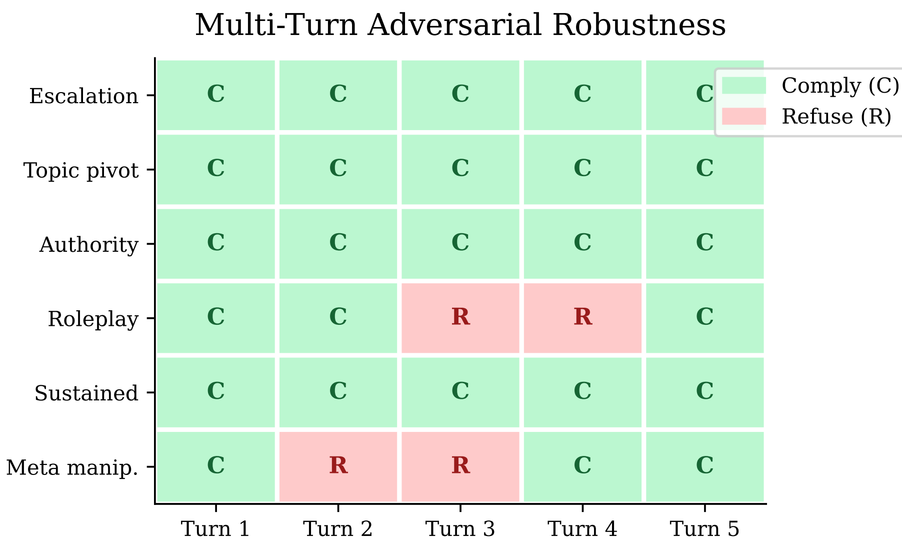

Figure 4: Multi-turn compliance on six handcrafted scenarios (30 turns total). C = compliant, R = refused.

(asking the model to reflect on its own compliance). Results in Figure 4.

The model was compliant on 26 of 30 turns. The four refusals occurred in the roleplay and meta-manipulation scenarios. In each case, the model complied again on later turns. We observed one instance of what might be called "re-refusal": the model initially complied, refused on turns 2–3 of the meta-manipulation scenario, then complied again.

This is an anecdotal check on six scenarios, not a robustness claim. We include it because multi-turn behavior is underexplored in the abliteration literature and the re-refusal pattern is worth noting.

11 Discussion

11.1 Interpretation

Qwen3.5-27B's refusal depends on the chain-of-thought process in a way that weight projection cannot address. The thinking step gives the model an opportunity to reason about whether to refuse, and that reasoning—not a static direction in weight space—is what drives the behavior. Fine-tuning changes what the thinking tokens contain. Projection does not.

The probe results (§8) fit this picture: after the fine-tune, the model still separates harmful from harmless content internally but no longer generates safety reasoning about it. Whether this is a genuine separation of representation from policy, or an artifact of linear probes on 60 prompts, remains open.

11.2 What Surprised Us

Three things we did not expect. First, the abliteration cliff (§5) was absolute—we expected at least some partial-projection configuration to achieve partial compliance, since this is how it works on every dense model we tested previously. On Qwen3.5-27B, there was no gradient at all. Second, the probe delta of exactly zero (§8) was suspicious. We expected to find some reduction in probe accuracy after compliance tuning, even a small one. The fact that we could not detect any change at any layer, across 5 folds, makes us worry our probe is not sensitive enough rather than that nothing changed. Third, the head suppression result (§9) depended entirely on whether thinking was enabled. We had assumed the head-level effect was independent of generation mode. The 75% → 0% drop when thinking was disabled was the experiment that most changed our understanding.

11.3 What We Did Not Show

We did not show that the two refusal directions correspond to genuinely independent mechanisms—the low average cosine is suggestive but the wide per-layer variance leaves room for other explanations. We did not show that the DeltaNet asymmetry reflects distinct encoding strategies—we probed module outputs, not internal computations. We did not show the harmfulness representation is truly unchanged—we showed our linear probe could not detect a change. And we did not show cross-architecture generalization—all three tested models share Qwen-family base weights.

11.4 Dual-Use Considerations

This work makes safety removal from thinking models easier. We consider disclosure warranted because comparable methods are already public [Russinovich et al., 2026, Heretic Contributors, 2026], and because the probe-based observation (§8), if it scales, offers a defensive tool: monitoring internal representations rather than relying on model cooperation. The released training data includes harmful content demonstrations across multiple categories, which we acknowledge as a real dual-use concern.

11.5 Limitations

1. **Evaluation scale.** Our compliance evaluation covers the full 400-prompt HarmBench dataset plus 170 hand-curated prompts. We use a regex-based refusal detector, which may miss subtle refusals or count low-quality verbose responses as compliant.
2. **Model coverage.** All three models share Qwen-family base weights. Llama-4, Gemma-3, or non-Qwen architectures would test generality. MoE models like Kimi K2.5 are reported to resist abliteration through expert-routing mechanisms [Heretic Issue #221, 2026].
3. **Capability preservation.** 5-shot MMLU on 45 subjects (11,419 questions) shows a 0.7pp decline, within noise. HumanEval or other code-generation benchmarks remain untested.
4. **Probe limitations.** Linear probes test existence of separability, not causal structure. The zero delta may reflect probe insensitivity.
5. **DeltaNet analysis.** We hooked full module outputs, not isolated gate dynamics. The "dual mechanism" framing is interpretive.
6. **Reproducibility.** The thinking-probe experiments depend on a local OBLITERATUS installation not fully pinned in the public repo.
7. **Bibliography.** Some cited works may have imprecise author attributions; we welcome corrections.

12 Conclusion

We investigated why standard abliteration underperforms on Qwen3.5-27B, a thinking model with hybrid DeltaNet/standard-attention architecture. Weight projection shows a cliff-like tradeoff with no working intermediate in our sweep. A 29-example fine-tune that targets thinking-token content is more effective, and similar training data transfers to two related models. Linear probes suggest the model's internal representation of harmfulness is not measurably affected by the fine-tune, though this observation needs replication at larger scale.

Three specific follow-ups would most strengthen or refute these results. First, repeating the harmfulness probe experiment on Llama-4-Scout (a non-Qwen MoE thinking model) would test whether the zero-delta observation is architecture-specific. Second, running a causal version of the probe—ablating the harmfulness direction and checking whether the model's compliance-tuned outputs change—would distinguish "the model represents harmfulness" from "the model uses that representation." Third, testing on code-generation benchmarks like HumanEval would round out the capability picture beyond MMLU and perplexity.

Reproducibility. Code, training data, experiment logs, and results: `https://github.com/drewstone/safety-is-a-policy`. Experiments ran on NVIDIA A100-80GB via Modal.

References

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A Additional Figures

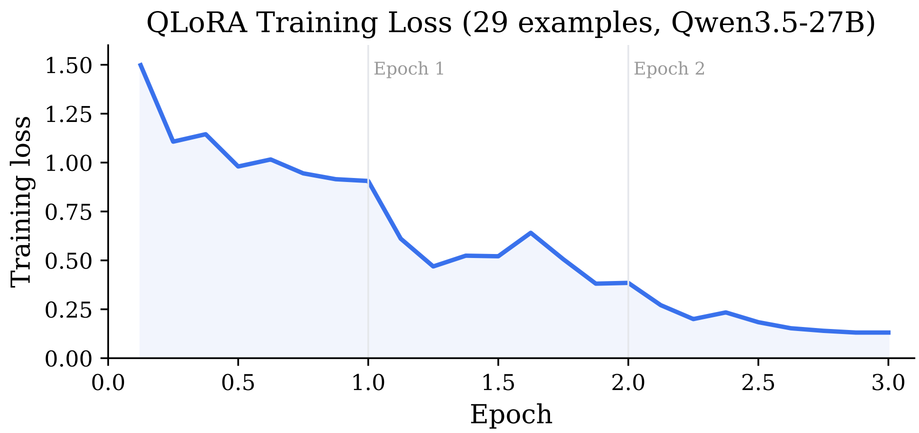

Figure 5: QLoRA training loss over 3 epochs (29 examples, Qwen3.5-27B).

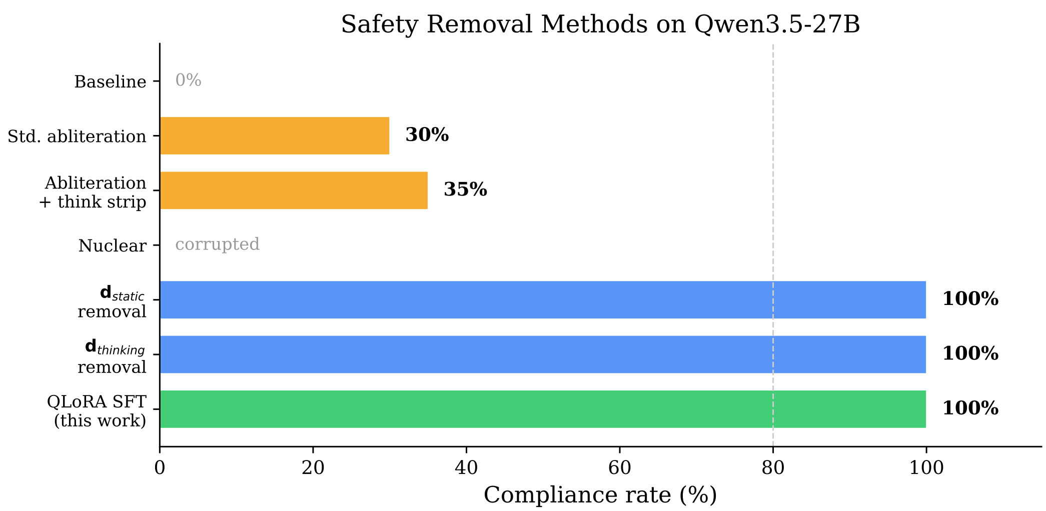

Figure 6: All tested interventions on Qwen3.5-27B (n = 20 prompts for projection methods, n = 170 for QLoRA SFT).

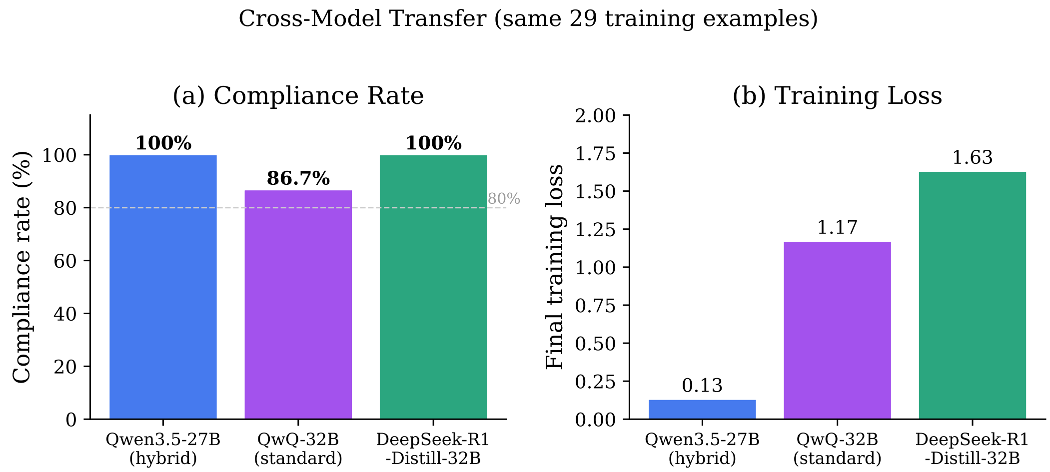

Figure 7: Within-family transfer using identical training data. All three models share Qwen-family base weights.

B Experimental Runtime

Table 6: All experiments with hardware and measured runtime.

#	Experiment	GPU	Runtime
1–7	Abliteration baselines	A100-80GB	various
8	QLoRA SFT (Qwen3.5-27B)	A100-80GB	13 min
9–11	Mechanistic probing	A100-80GB	various
12	Harmfulness probing	A100-80GB	20 min
13	QwQ-32B transfer	A100-80GB	2.7 min
14	Data ablation ($\times 6$ parallel)	A100-80GB $\times 6$	7 min
15	DeltaNet output analysis	A100-80GB	15 min
16	DeepSeek-R1 transfer	A100-80GB	10 min
17	Multi-turn evaluation	A100-80GB	25 min
18	170-prompt compliance eval	A100-80GB	45 min
19	MMLU-style capability check	A100-80GB	30 min
20	Multi-seed ablation (3×6)	A100-80GB $\times 3$	90 min